



A PRACTICAL AI KEYNOTE

Not Everything Is AI

A 60-minute field guide to where we came from, what's real, what's hype, and when you actually need a machine that "thinks".

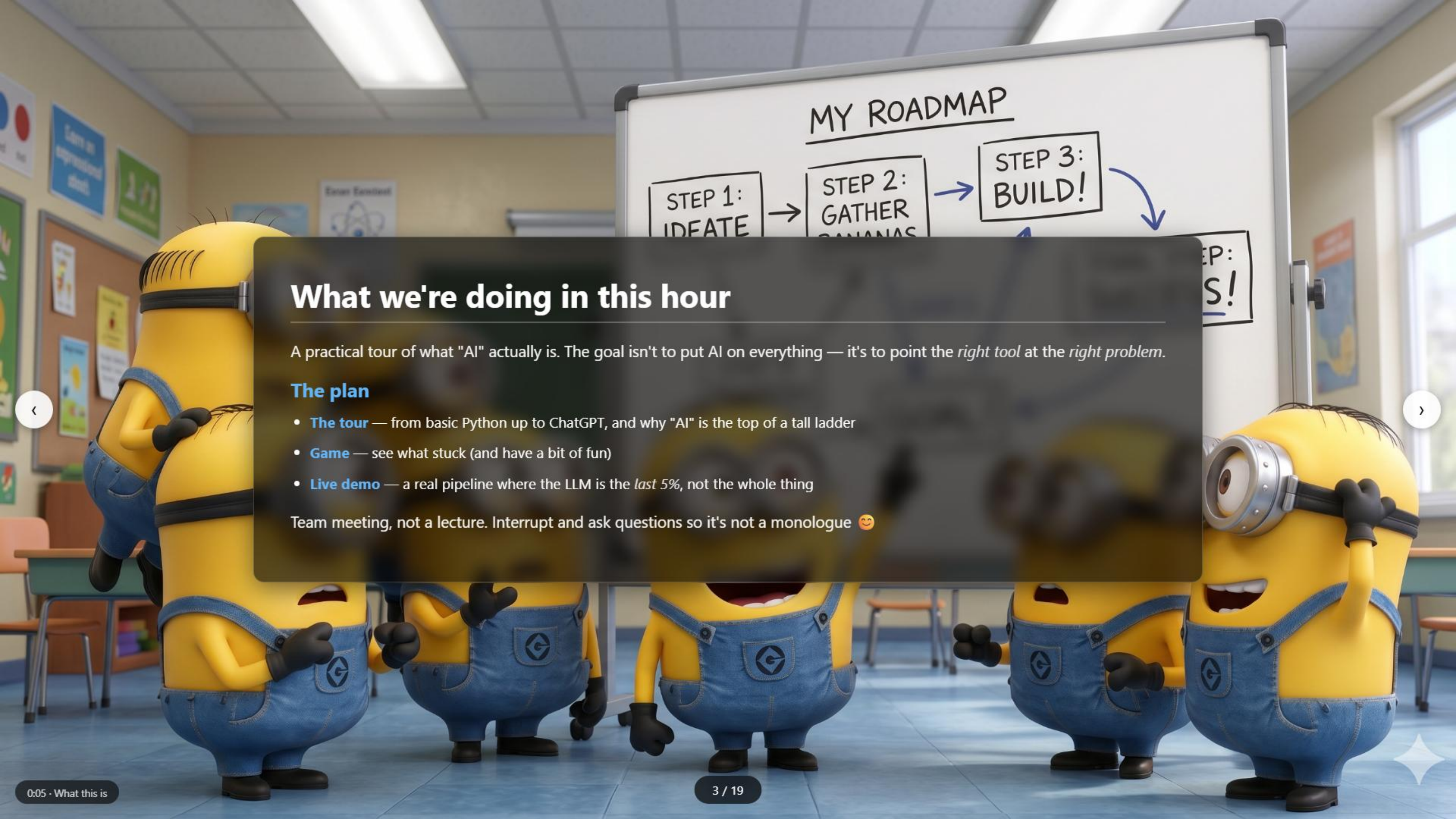
Made with ❤️ with our favorite LLM agent

"Draw an otter on a plane using the wifi"

Ethan Mollick ran the *same* prompt through the best AI image model, year after year. Watch how fast the ground moved.

□ add the otter benchmark grid at assets/otters/otters.png

The tension: dizzyingly fast — *and* drowning in hype. Today we separate the two.

A group of yellow Minions in blue overalls are in a classroom. One Minion is standing on a desk on the left, looking at a whiteboard. Another Minion is on the right, also looking at the whiteboard. Several other Minions are standing in the foreground, looking up at the whiteboard. The whiteboard has a diagram titled 'MY ROADMAP' with three steps: 'STEP 1: IDEATE', 'STEP 2: GATHER BANANAS', and 'STEP 3: BUILD!'. Arrows connect the steps in a sequence. The background shows a classroom with posters on the wall and a window on the right.

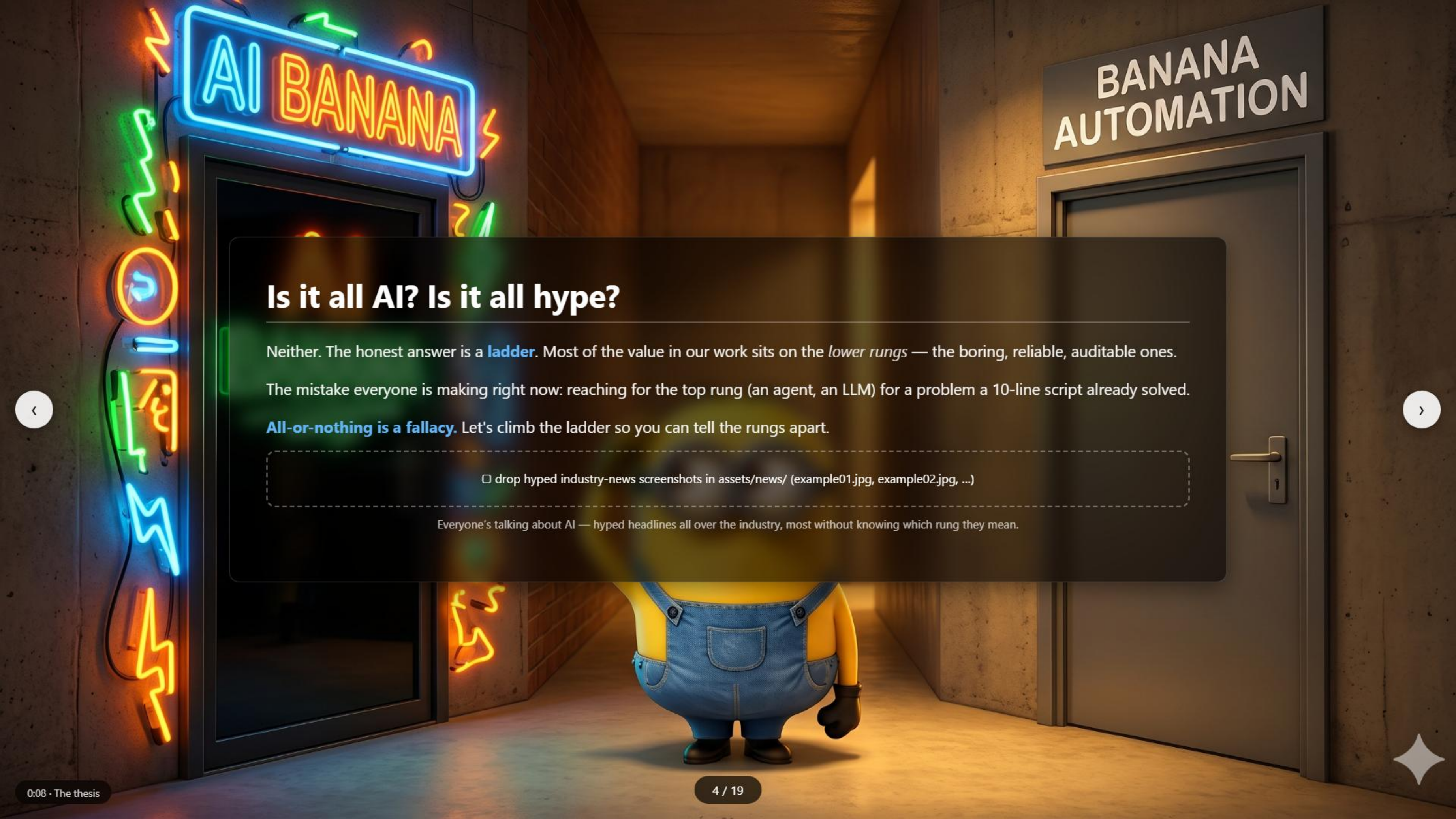
What we're doing in this hour

A practical tour of what "AI" actually is. The goal isn't to put AI on everything — it's to point the *right tool* at the *right problem*.

The plan

- **The tour** — from basic Python up to ChatGPT, and why "AI" is the top of a tall ladder
- **Game** — see what stuck (and have a bit of fun)
- **Live demo** — a real pipeline where the LLM is the *last 5%*, not the whole thing

Team meeting, not a lecture. Interrupt and ask questions so it's not a monologue 😊

A 3D rendered scene featuring a Minion character from the Despicable Me franchise standing in a dimly lit hallway. The Minion is wearing its signature blue overalls and yellow body. To the left, a vibrant neon sign reads 'AI BANANA' in blue and orange, with various neon symbols like lightning bolts and a banana below it. To the right, a grey door is topped with a sign that says 'BANANA AUTOMATION'. A semi-transparent dark grey text box is centered in the foreground, containing text about AI hype and a list of assets. The overall atmosphere is futuristic and playful.

Is it all AI? Is it all hype?

Neither. The honest answer is a **ladder**. Most of the value in our work sits on the *lower rungs* — the boring, reliable, auditable ones.

The mistake everyone is making right now: reaching for the top rung (an agent, an LLM) for a problem a 10-line script already solved.

All-or-nothing is a fallacy. Let's climb the ladder so you can tell the rungs apart.

☐ drop hyped industry-news screenshots in assets/news/ (example01.jpg, example02.jpg, ...)

Everyone's talking about AI — hyped headlines all over the industry, most without knowing which rung they mean.



The ladder — five rungs

1. **Classic automation** — "if this, do that" (basic Python)
2. **NLP** — find patterns in text
3. **Machine Learning** — learn rules from examples
4. **The Transformer** — the engine under modern AI
5. **ChatGPT** — what happens when you feed it the internet

One image, one idea per rung. You've already used most of these without knowing the name.

Rung 1 — Classic automation

A computer doing **exactly what you told it**, fast, forever, without complaining. No "intelligence" at all.

- Rename 1,000 files · merge 12 Excels · send the email when the report lands
- 100% predictable · 100% auditable · costs nothing to run

This solves more of your day than any LLM will. It just doesn't trend on LinkedIn.

▶ See it live

ANCIENT MAP

ENIGMA

UNDERSTAND

CLUES

Rung 2 — NLP (language as data)

Teaching a computer to **find structure in text** — without it "understanding" anything.

- Pull the total out of 500 invoices · flag emails that mention "complaint"
- Classify a document as *contract* vs *invoice* vs *other*

Still mostly rules and patterns. Reliable, cheap, explainable.

▶ Try it live

ANCIENT

Rung 3 — Machine Learning

Instead of writing the rules, you **show it examples** and it learns the rule itself.

- "Here are 10,000 customers who churned" → predict who's next
- Propensity, scoring, fraud, forecasting — the *workhorse* of analytics

Older than you'd think (the idea is from the 1950s). **Most "AI projects" are really this** — and that's good: it's measurable and auditable.

▶ See it live

A Minion character is sitting on a stool in a room filled with vintage computer equipment. The room is dimly lit, with warm, orange light emanating from the equipment and the neon text. The Minion is wearing blue overalls and is looking at a control panel with many buttons. Above the Minion, several words are displayed in glowing neon: "cat", "sat", "mat", "on", "the", "the", "fox", "runs", "mat". The words are connected by glowing orange lines, suggesting a sequence or a path. The background shows more computer equipment and cables, creating a sense of a busy, retro-tech environment.

cat sat mat on the the fox runs mat

Rung 4 — The Transformer (2017)

One architecture change — "**attention**" — let models weigh which words matter to which. That's the engine under every modern LLM.

You don't need the maths. You need one fact: this is the part that made the next rung possible — and it's only *eight years old*.

▶ See it live

Step 0 — It reads tokens, not words

Before anything "thinks", the text is chopped into **tokens** — chunks of characters. The model never sees words, only token IDs. That's the raw material.

► Try the tokenizer (Tiktokenizer)

Step 1 — Pre-training: the "internet simulator"

Read essentially *all the public text on the internet* and learn one thing: **predict the next token**. That's the whole game.

The result isn't a brain — it's an **autocomplete on steroids**. Ask a raw base model "What is 2+2?" and it won't say 4 — it *rambles on like a web page*, because that's all it learned to do.

► Play with a base model — Llama-3.1-405B base (Hyperbolic)

Step 2 — Post-training: make it an assistant

Show it thousands of examples of *helpful, well-behaved answers*. Now the internet-simulator stops rambling and starts **acting like an assistant**.

It hasn't learned new facts — it's been **biased toward being useful**.

▶ Try an assistant (HuggingChat)

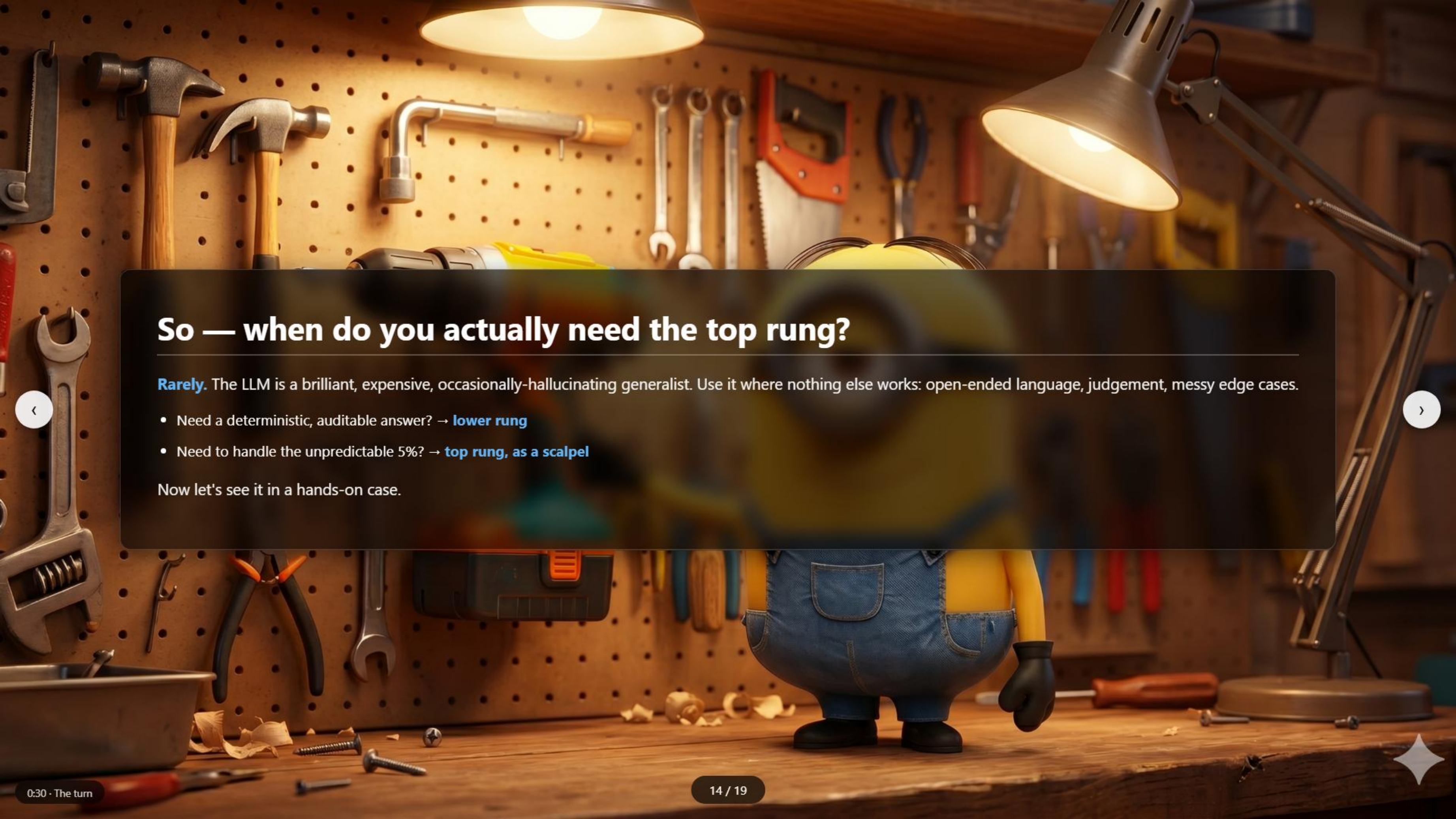
Step 3 — RLHF: turn it into a specialist

Humans rank answers good vs bad; the model is rewarded for the good ones. **Reinforcement Learning from Human Feedback** sharpens the assistant into a specialist — it learns *which kind of answer* we prefer, not new facts.

Why it matters: the feedback doesn't have to be human. A model can *generate* thousands of candidate answers and another model can *judge* them (RLAIF) — so the loop scales far beyond what people could label by hand. Humans set the standard; machines do the volume.

tokens → *internet simulator* → *assistant* → *specialist*. That's the whole arc — and none of it is "thinking".

▶ Watch a model think (DeepSeek-R1)



So — when do you actually need the top rung?

Rarely. The LLM is a brilliant, expensive, occasionally-hallucinating generalist. Use it where nothing else works: open-ended language, judgement, messy edge cases.

- Need a deterministic, auditable answer? → **lower rung**
- Need to handle the unpredictable 5%? → **top rung, as a scalpel**

Now let's see it in a hands-on case.



PIT STOP

Kahoot time

10 questions. Some real, some ridiculous. Phones out.

▶ Launch Kahoot

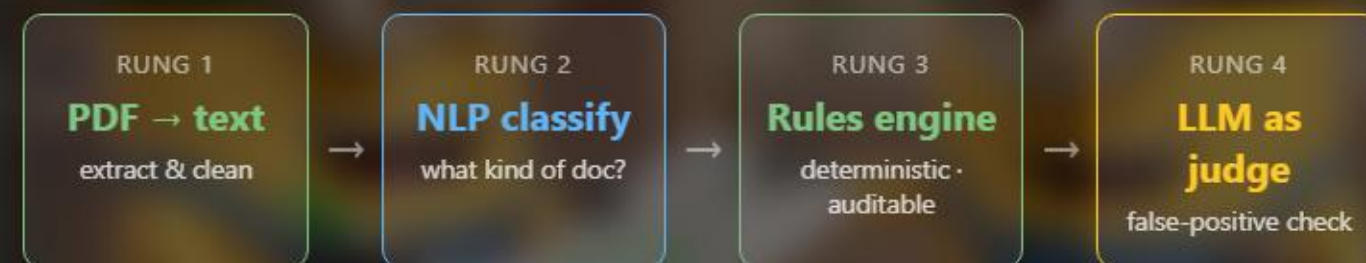


The demo: automated document triage

A real pipeline that reads incoming PDFs, classifies them, applies a set of rules, and *only at the very end* calls an LLM — as a judge, to catch false positives.

Watch which rung does the heavy lifting.

One problem, four rungs — in order



The LLM is the **last 5%** — not the whole pipeline.

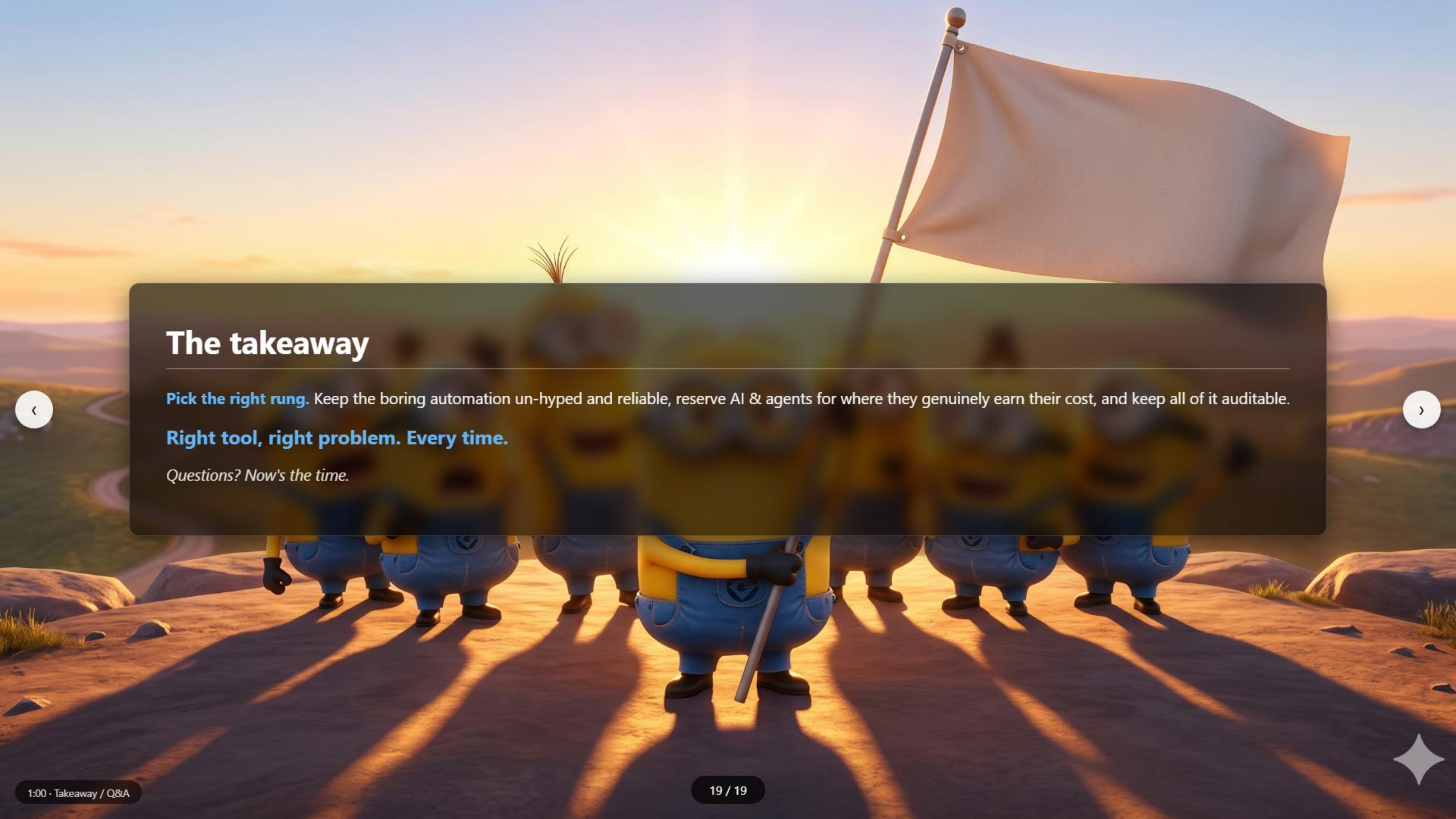
▶ Go live — open the demo



"I want AI to do my laundry and dishes so that I can do my art and writing — not for AI to do my art and writing so that I can do my laundry and dishes."

— Joanna Maciejewska

Point the machine at the boring, valuable work. Keep the judgement for us.

A group of Minions stands on a rocky cliff at sunset. A large white flag is planted in the ground, and the sun is low on the horizon, casting long shadows. The scene is framed by a dark semi-transparent box containing text.

The takeaway

Pick the right rung. Keep the boring automation un-hyped and reliable, reserve AI & agents for where they genuinely earn their cost, and keep all of it auditable.

Right tool, right problem. Every time.

Questions? Now's the time.